Status	Finished
Started	Thursday, 20 August 2026, 2:03 PM
Completed	Thursday, 20 August 2026, 2:12 PM
Duration	8 mins 39 secs
Grade	**9.00** out of 10.00 (**90**%)

Question 1

Complete

Mark 1.00 out of 1.00

Which statement best describes the main function of an oscilloscope?

- a. It converts a digital data stream directly into an analog message.
- b. It measures only the total bandwidth available in a network channel.
- c. It calculates the IP address assigned to a communication interface.
- d. It determines the routing path between two communicating computers.
- e. It displays variations of an electrical signal as a function of time.

The correct answer is: It displays variations of an electrical signal as a function of time.

Question 2

Complete

Mark 1.00 out of 1.00

What is meant by sampling a signal?

- a. Removing all high-frequency components from a transmitted signal
- b. Converting several signals into one frequency-domain component
- c. Sending every possible signal value continuously through a channel
- d. Increasing the amplitude of a signal before its transmission
- e. Measuring the signal value at selected points in time

The correct answer is: Measuring the signal value at selected points in time

Question 3

Complete

Mark 1.00 out of 1.00

What can happen when the sampling rate is too low or aliasing effect?

- a. A high-frequency component may be interpreted as a lower-frequency component.
- b. The reconstructed frequency increases automatically as the sampling rate decreases.
- c. The original frequency remains unchanged provided that enough samples are stored.
- d. The sampled signal always retains the original frequency but loses amplitude.
- e. The sampling process removes high-frequency components without affecting reconstruction.

The correct answer is: A high-frequency component may be interpreted as a lower-frequency component.

Question 4

Complete

Mark 1.00 out of 1.00

A signal has frequency components from 2 kHz to 15 kHz.

In order to avoid aliasing when digitizing this signal, the **minimum sampling rate** is [30] kHz

Hint: Slide 13

fs >= 2*fmax = 2(15) = 30 kHz

Question 5

Complete

Mark 1.00 out of 1.00

A periodic signal consists of components at 200 Hz, 400 Hz, 600 Hz, and 1000 Hz.

This signal's bandwidth = [800] Hz

B=f_high−f_low=1000−200=800 Hz

Question 6

Complete

Mark 0.00 out of 1.00

A digital signal has a bit rate of 4000 bps.

The duration of one bit (the bit interval) = [4] µs

Question 7

Complete

Mark 1.00 out of 1.00

If a system uses 16 distinct signal levels, then the number of bits per symbol = 4 bits.

Question 8

Complete

Mark 1.00 out of 1.00

What is the communication-design goal described when comparing data rate and signal rate?

a. Reduce data rate while increasing the number of transmitted symbols.

b. Keep both data rate and signal rate equal to zero during transmission.

c. Increase signal rate regardless of the amount of transmitted information.

d. Keep the number of signal levels fixed at two for every system.

e. Increase data rate while reducing the required signal rate when possible.

The correct answer is: Increase data rate while reducing the required signal rate when possible.

Question 9

Complete

Mark 1.00 out of 1.00

A signal carries data in which **one signal element represents two data elements** (r=2).

If the bit rate N is **120 kbps** and c is 0.5, then calculate the average signal rate: = 30 kbaud.

Hint: Slide 34

Question 10

Complete

Mark 1.00 out of 1.00

A digital signal uses **8 signal levels**. If each signal element carries the maximum possible number of bits and the bit rate is **240 kbps**, what is the average signal rate for c=0.5?

Hint: Slide 32 and 34

Answer: 40 kbaud

$S = c * N/r = 0.5\ (240)/3$

$r = \log_2 8 = 3$